



ARTICLE



<https://doi.org/10.1057/s41599-025-05924-3>

OPEN

Large Language Models in Legal Systems: A Survey

Fatemeh Dehghani¹, Roya Dehghani¹, Yazdan Naderzadeh Ardebili² & Shahryar Rahnamayan³

This paper provides a comprehensive survey of the role of large language models (LLMs) in legal systems. It examines their applications across key areas such as legal document drafting, case analysis, research, compliance monitoring, and education. In addition to mapping these use cases, the survey reviews datasets and benchmarks that enable the training and fine-tuning of LLMs for legal tasks. The analysis highlights both the opportunities and challenges of adopting LLMs in practice, including issues of bias, interpretability, accuracy, and ethical risk. Particular attention is given to the limitations of current models and the risks of overstating their reliability in high-stakes legal contexts. By synthesizing recent advancements, this paper provides a balanced perspective on the current state of LLMs in the legal domain and outlines future directions for research and practice aimed at improving their effectiveness, accountability, and responsible deployment.

¹University of Ontario Institute of Technology, Oshawa, Canada. ²University of Toronto, Toronto, Canada. ³Brock University, St. Catharines, Canada.
email: fatemeh.dehghani@ontariotechu.net

Introduction

LLMs are Artificial Intelligence (AI)-based models that excel at understanding and creating human language, closely matching actual human conversation. These models use deep learning techniques and are trained on massive volumes of diverse and high-quality text data, allowing them to thoroughly capture language structures, grammar, context, and semantics. *GPT-3*, *GPT-4*, *LaMDA*, *BLOOM*, and *LLaMA* are all instances of LLMs. These models have changed a wide range of daily applications, including but not limited to chatbots, AI search engines, and content generating tools. *GPT-3*, for example, created by *OpenAI*, made major advances in Natural Language Processing (NLP). It displayed exceptional language creation skills with its remarkable 175 billion parameters, paving the path for later models (SuperAnnotate, 2023).

While GPT-based models like *ChatGPT* have dominated public discourse, other foundational LLMs are gaining traction across domains, including legal services. *LaMDA* (Language Model for Dialogue Applications), developed by Google, focuses on nuanced, open-ended dialogue and holds potential in client-facing legal tools. *LLaMA* (Large Language Model Meta AI), created by Meta, provides open-access models that have been adapted into specialized variants such as *Lawyer LLaMA*, designed to improve legal reasoning and document analysis (Huang et al., 2023). Meanwhile, *BLOOM*, developed through the collaborative BigScience project, supports over 40 languages and has been explored for multilingual legal research and compliance across international jurisdictions (Workshop et al., 2022). These models broaden the scope of LLM deployment and offer complementary capabilities beyond those of GPT-based systems.

LLMs have significantly altered the landscape of NLP and opened up new possibilities in fields such as reinforcement learning, NLP, and computer vision (CV) integration. Researchers have investigated the use of LLMs in conjunction with reinforcement learning to assess and guide the learning process, resulting in better performance and quicker convergence (Du et al., 2023). LLMs have played a leading role in NLP research, enabling progress in areas like as language modeling, machine translation, sentiment analysis, and question-answering systems (Brown et al., 2020; Touvron et al., 2023). Furthermore, they have been employed in computer vision tasks to stimulate reasoning and produce descriptions for visual content (Shao et al., 2023; Wu et al., 2023a).

Given their success in general-purpose language tasks, researchers and practitioners have increasingly turned to LLMs to explore their potential in the legal domain.

The ability of LLMs to enhance efficiency and improve legal practitioners' production has shown great promise in the legal industry. Despite privacy and data confidentiality issues, several law firms have started testing LLMs for a variety of use cases, including document preparation, case law research, and competitive analysis. Integrating generative AI has emerged as a game changer, allowing law companies to improve and simplify their legal services.

One notable example of a generative AI product that has gained significant popularity is *ChatGPT*. This powerful linguistic model achieved worldwide recognition at an unprecedented rate. According to *OpenAI*, five days after its launch in November 2022, *ChatGPT* had already attracted one million users. Within just two months, it reached an extraordinary milestone of 100 million users. Remarkably, *ChatGPT* surpassed the growth rates of well-known online platforms such as Facebook, Instagram, and Spotify. This rapid adoption underscores the growing demand and public enthusiasm for conversational AI technologies. Figure 1 (Duarte, 2023) illustrates

how quickly *ChatGPT* expanded in comparison to other popular online services, measured by the time taken to reach one million users. Figure 2 shows the current number of these online services in 2024 (SEO.ai, 2024; Backlinko, 2024; Dropbox, 2024; Kickstarter, 2024; Foursquare, 2024); Statista, 2024).

The use of LLMs in the legal sector has the potential to transform several elements of legal processes. With the use of LLMs, document preparation, a generally time-consuming activity, may be greatly accelerated and made more efficient. These algorithms can evaluate large quantities of legal text, extract relevant information, and create correct and personalized legal documents, saving legal practitioners considerable time and resources.

Case law study is another area where LLMs have proven their worth. These algorithms can effectively filter through large datasets, find important precedent cases, and give legal practitioners with complete summaries and insights. Law companies can use LLMs to speed up the research process, allowing lawyers to make better informed judgments based on a deeper grasp of legal precedents. Furthermore, LLMs have shown to be effective instruments for legal competitiveness analysis. These algorithms may discover patterns, trends, and possible legal issues by analyzing enormous amounts of data, such as court cases, regulatory filings, and market reports. Law firms may gain a competitive advantage by leveraging the capabilities of LLMs to keep updated about legal changes, analyze market trends, and make data-driven strategic choices.

According to the report (Warren, 2023), 80% of law firm Chief Executives feel that generative AI can be used to legal practice, with more than 50% believing that it should be. According to a breakthrough *LexisNexis* survey (LexisNexis, 2023), 39% of lawyers, 46% of law students, and 45% of customers expect generative AI technologies to significantly revolutionize the practice of law. The survey, which included 4,180 participants, found a considerable awareness gap regarding generative AI between the legal community (88%) and the general population (57%). While 57% of customers are aware of *ChatGPT*, just 32% have used it, with 15% seeking legal guidance. With 86% awareness, lawyers have a significant conviction (84%) in the efficiency improvement that generative AI may give.

Despite a generally favorable perception (77% positive or mixed), ethical concerns about generative AI remain widespread—87% of lawyers, 91% of law students, and 72% of consumers express concerns. Meanwhile, 61% of lawyers and 44% of law students believe that generative AI will revolutionize law schools and the way law is taught and studied. Figure 3 shows the uses of generative AI tools in their daily work tasks. General legal advice (60%), drafting a will (43%), and legal requirements for company establishment (41%), are the top topics consumers evaluate for AI usage, offering insight on the growing landscape of AI in the legal profession, which is shown in Fig. 4.

LLMs in Legal Services: Applications and Innovations

LLMs have significantly reshaped the legal services landscape by enabling automation, enhancing decision-making, and improving accessibility across various legal domains. This section explores four key areas where LLMs are making substantial contributions: (1) legal education and training, (2) legal practice, (3) legal reasoning and judgment prediction, and (4) legal text categorization and summarization. Each subsection outlines the current advancements, practical implementations, and supporting research that illustrate how LLMs are transforming the legal profession.

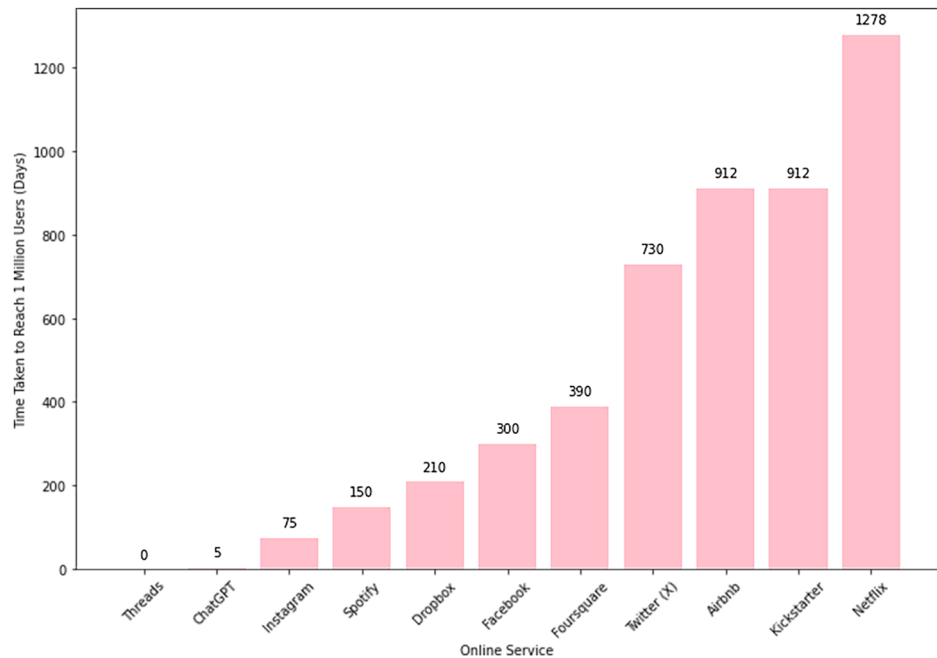


Fig. 1 Time taken for selected online services to reach one million users (Duarte, 2023).

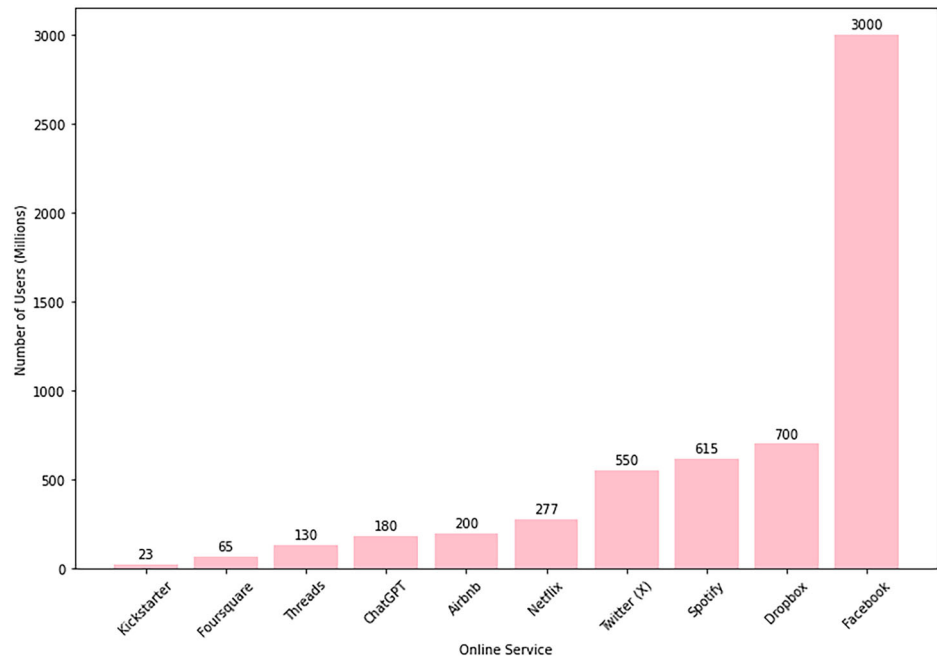


Fig. 2 Current number of users for various online services in 2024.

A. Applications in Legal Education and Training. LLMs, such as *ChatGPT*, have shown considerable promise in the legal education domain. According to (Choi et al., 2021), *ChatGPT* can support legal writing by assisting with drafting documents, helping users get started when they experience writer’s block (i.e., difficulty beginning or continuing their writing), and enhancing the quality and speed of legal argument development. Additionally, it may aid students in reciting legal rules and synthesizing complex case law, allowing them to better apply those rules to specific legal contexts.

In a similar context, (Oltz, 2023) explored how *ChatGPT* might ease the teaching workload of law professors. The study tested the model’s ability to generate exam questions, syllabus, and

recommendation letters. Results suggest that these tools can boost instructors’ creativity and productivity by providing reusable and customizable templates.

An additional educational use case, highlighted in (Armstrong, 2023), involves using *ChatGPT* to identify logical errors in contract clauses. This feature serves as an instructional aid for teaching contract law and improving students’ analytical skills.

Recent evaluations of LLMs such as *GPT-3.5* and *GPT-4* also highlight their capacity to undertake legal examination tasks. *GPT-3.5* scored 50.3% on a practice version of the National Conference of Bar Examiners’ Multistate Bar Examination (NCBE MBE)—a standardized multiple-choice test that assesses core legal knowledge across subjects such as Constitutional

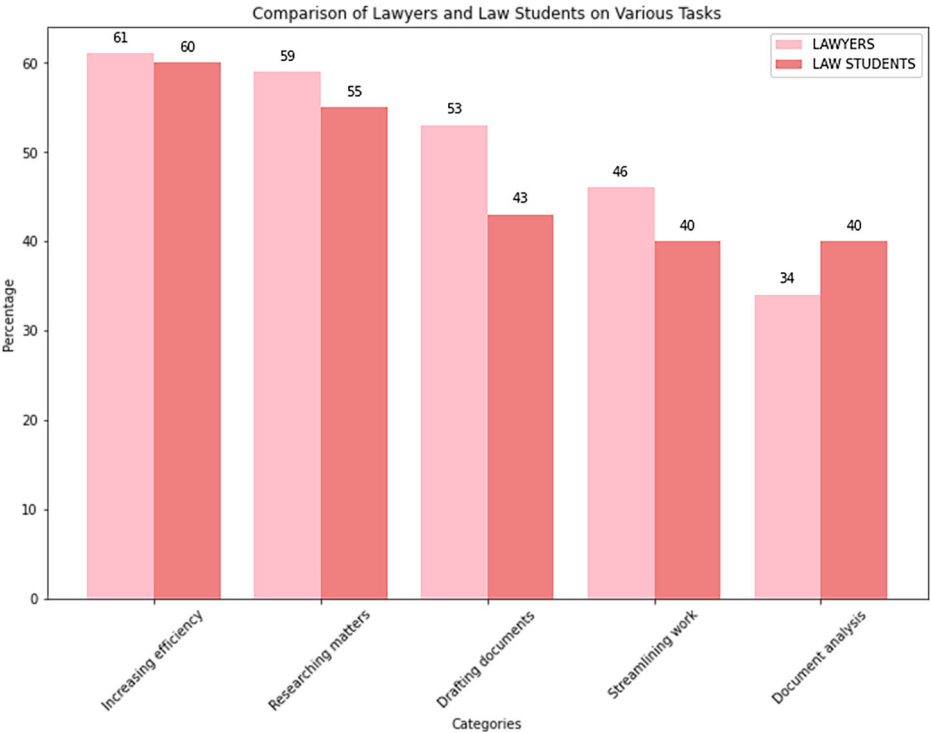


Fig. 3 The uses of generative AI tools by lawyers and law students in their daily work (LexisNexis, 2023).

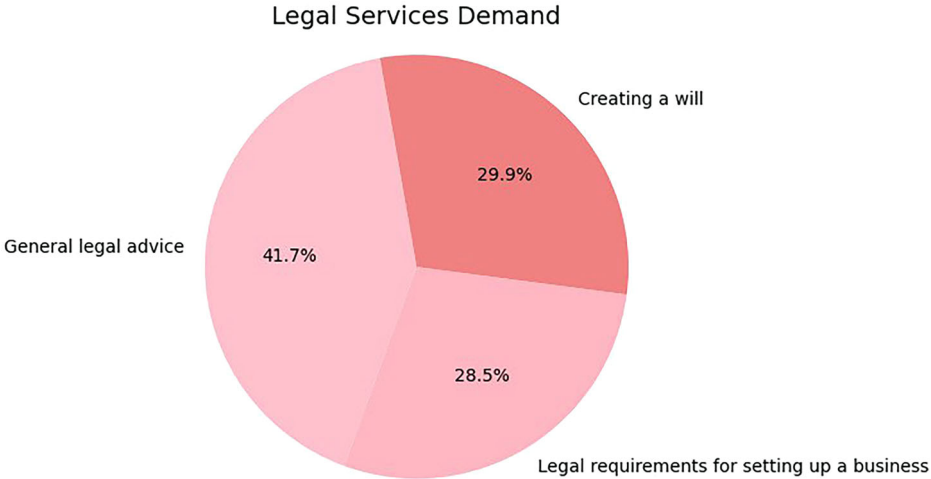


Fig. 4 The top three legal areas that users employ generative AI (LexisNexis, 2023).

Law, Contracts, and Criminal Procedure—excelling particularly in Evidence and Torts (Bommarito and Katz, 2022). *GPT-4* further improved on this, achieving an average score of 4.2/6.0 on the Multistate Essay Exam (MEE) and Multistate Performance Test (MPT), surpassing the passing threshold in all Uniform Bar Examination (UBE) jurisdictions (Katz et al., 2023).

B. Applications in Legal Practice. LLMs also provide significant enhancements to various aspects of legal practice. Below, we categorize these contributions into distinct functional areas.

1. *Document Management and Drafting:* LLMs can automate the drafting of legal documents such as wills, trusts, contracts, and client communications (Macey-Dare, 2023). They can also assist in managing case-related

documents, monitoring contract deadlines, and organizing materials for case preparation. These capabilities reduce the workload and turnaround time for legal professionals (Parizi et al., 2023). More advanced systems, like *ChatLaw* (Cui et al., 2023) and *Lawyer LLaMA* (Huang et al., 2023), further demonstrate the potential of domain-specific LLMs in automating legal documentation with higher precision.

2. *Legal Analysis and Decision Support:* LLMs can analyze legal data to support case strategy, risk assessment, and outcome prediction. By identifying trends and referencing past case outcomes, they can inform legal decision-making with a degree of predictive accuracy. Legal analytics tools have demonstrated approximately 70% accuracy in predicting case outcomes using historical data (Niklaus et al., 2021). GPT-based tools have also been used in extracting

structured information from employment tribunal judgments, demonstrating their potential for knowledge extraction from unstructured legal texts (de Faria et al., 2024).

3. *Regulatory Compliance and Due Diligence*: LLMs assist legal professionals in complying with complex regulatory frameworks and identifying potential legal violations. Tasks such as fraud detection, employee vetting, and compliance reviews are increasingly supported by these tools. These models are particularly useful for reviewing financial records and detecting red flags during due diligence processes. A recent overview by Aimultiple also highlights the integration of generative AI into various due diligence workflows across legal domains (Team, 2025).
4. *Client Engagement and Operational Support*: AI-powered chatbots are being used by legal firms to handle client queries related to billing, scheduling, and case updates. Additionally, virtual legal assistants help with managing calendars and to-do lists, and even provide preliminary legal advice or facilitate alternative dispute resolution processes like arbitration and mediation (Macey-Dare, 2023). However, as noted by Stanford HAI (The Stanford Institute for Human-Centered AI), performance inconsistencies and hallucination risks in LLM-generated legal advice still warrant caution and proper oversight (Magesh et al., 2024).
5. *Legal Research and Knowledge Retrieval*: Legal research is a key application of LLMs in practice. These systems expedite research by retrieving case laws, interpreting statutes, and generating legal briefs and memos. GPT-based models have been tested on real-world legal codes (e.g., the Polish Penal Code) and achieved promising classification results (Weichbroth et al., 2025). The growing body of work, such as the Harvard CLP article on GPT-3 in legal services, reflects the importance of continued study into responsible and accurate AI use in legal settings (Perlman, 2023; Zang et al., 2024).

Recent advancements have further expanded the scope of LLM-based legal practice through the integration of reasoning modules, retrieval augmentation, and agent-based interaction systems. DISC-LawLLM (Yue et al., 2023), for example, combines fine-tuned LLMs with external legal databases to improve factual accuracy and context awareness in legal research and advisory tasks. Similarly, the latest version of ChatLaw (Cui et al., 2023) incorporates multi-agent collaboration and knowledge graphs to deliver structured, jurisdiction-aware legal responses, reducing hallucination and enhancing consultation reliability. These innovations signal a broader movement beyond GPT-based systems, toward hybrid architectures that emphasize transparency, domain grounding, and adaptability across legal subfields. Complementing this trend, recent legal NLP surveys (Qin and Sun, 2024) highlight increasing interest in multilingual models and jurisdiction-specific pipelines, pointing to the relevance of open-source models like BLOOM and LLaMA in diverse legal systems.

C. Legal Reasoning and Judgment Prediction. Blair et al. (Blair-Stanek et al., 2023) investigate the legal reasoning skills of the GPT-3 model, which is comparable to *ChatGPT*. Using a dynamic few-shot prompting strategy, which comprises presenting the model with restricted instances and an adaptive guidance technique, dramatically improved the model's precision and assurance in this scenario. Prompting method developments have had a substantial influence on the functionality of Language Models for Legal Reasoning in diverse legal projects.

Some researchers are using these models (LLMs) to examine fiduciary duties, as depicted in (Nay, 2023). This study uses

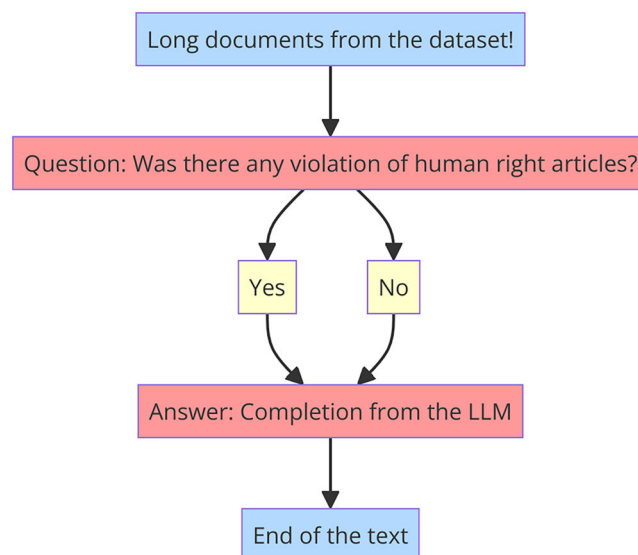


Fig. 5 An instance of a sample English prompt template for ECHR task (Trautmann et al., 2022).

natural language prompts taken from U.S. court cases to illustrate that LLMs can capture the “spirit” of direction or communication, allowing for improved understanding between legal terms or legal requirements and humans in AI agents. LLMs are used to illustrate how they understand the implied meanings that characterize fiduciary obligation, a difficult-to-interpret terminology by its nature. This competency is crucial when giving legal advice or making decisions related to fiduciary duties.

In (Trautmann et al., 2022), the authors propose Legal Prompt Engineering (LPE), a method of guiding and assisting an LLM in performing a Natural Legal Language Processing (NLLP) skill, to enhance the LLM's performance in tasks involving the prediction of legal judgments. Three multilingual datasets were used to test this approach, showcasing the model's capacity to manage the complexity of legal terms and thinking across several information sources. LPE entails customizing prompts for unique legal circumstances, allowing the LLM to better learn and analyze the subtleties of legal texts. For instance, research by Shui et al. demonstrates that incorporating similar cases and multi-choice options into LLM prompts significantly enhances the models' legal reasoning capabilities. By presenting LLMs with contextually relevant cases and structured options, the models can better recall domain-specific knowledge crucial for expert legal reasoning. This approach not only improves the accuracy of the LLMs' predictions but also their ability to handle complex legal queries effectively (Shui et al., 2023).

The effectiveness of zero-shot LPE for specific facts is examined in case texts from the Supreme Court of the Federal Republic of Switzerland (in German, French, and Italian) and the European Court of Human Rights (in English). According to their results, zero-shot LPE performs better than baselines but not as well as the most advanced supervised methods available today. The findings are noteworthy because no specific domain-specific information was utilized, indicating that general-purpose LLMs may migrate to the legal area without additional training or fine-tuning, saving enormously in terms of computational costs. Figure 5 shows an instance of a sample English prompt template for the European Court of Human Rights (ECHR) task.

LLMs have demonstrated promising performance in legal judgment prediction challenges. For example, a research utilizing the *CAIL2018* dataset to assess the ability of LLMs in predicting

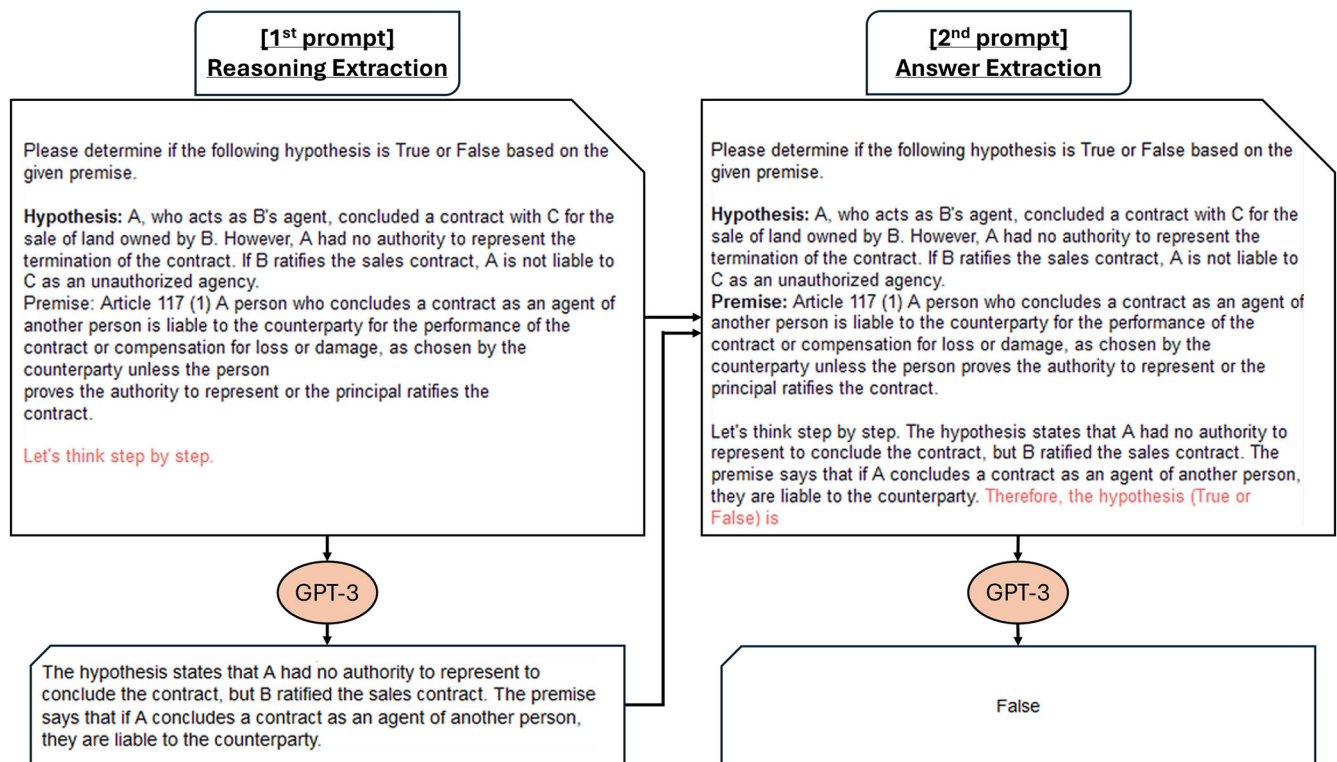


Fig. 6 Illustration of the Zero-Shot (ZS) Chain-of-Thought (CoT) process used for legal reasoning tasks. The first prompt (Reasoning Extraction) involves presenting GPT-3 with a hypothesis and assumption and asking it to determine the truth value based on logical reasoning. The model produces a detailed reasoning response. The second prompt (Answer Extraction) builds on this reasoning by asking GPT-3 to explicitly state whether the hypothesis is true or false. This process demonstrates how CoT prompts can guide the model to produce structured, logical analyses similar to the analytical processes used by lawyers (Yu et al., 2022).

judicial verdicts found accuracy rates of roughly 69% (Strickson and Iglesia, 2020). A notable advancement in this domain is the application of contrastive learning to improve model accuracy. Zhang et al. (Zhang et al., 2023) proposed a supervised contrastive learning framework specifically designed to help LLMs distinguish between closely related legal provisions. These provisions often fall under the same chapter of the criminal law—for instance, a chapter on property crimes might include distinct articles covering theft, fraud, and embezzlement. Although these charges belong to the same thematic category, each has unique legal definitions and consequences. The contrastive learning approach allows the model to learn fine-grained differences between such articles, enhancing its ability to assign the correct legal charge during judgment prediction. Additionally, Jiang and Yang (Jiang and Yang, 2023) introduced legal syllogism prompting (LoT), a method that teaches LLMs to use legal syllogism for judgment prediction. This technique enhances the model's focus on key information relevant to judgments and improves performance compared to other prompting methods.

Lastly, Chen et al. (Chen et al., 2022) explored the importance of understanding causality in Legal Judgment Prediction (LJP). They proposed methods to enhance model robustness and generalization by distinguishing non-causal information and applying causality-aware self-attention mechanisms. Their results showed that incorporating causality significantly improved the performance of LJP models.

These advancements demonstrate the growing capabilities of LLMs in LJP, making them valuable tools for enhancing legal decision-making processes.

D. Legal Text Categorization and Summarization. Using the *LexGLUE* benchmark, (Chalkidis, 2023) assess *OpenAI's* GPT-3.5

(*ChatGPT*) model for legal text categorization tasks. The micro-F1 score, a metric for evaluating classification models in multi-class or multi-label scenarios, is calculated by averaging the F1-scores of all classes (Takahashi et al., 2022). In these tasks, *ChatGPT* achieved an average micro-F1 score of 49.0%, which is better than simple guessing but significantly lower than that of smaller, more specialized models. Despite showing promise in several datasets, such as *LEDGAR* and *ECtHRB*, the study concludes that *ChatGPT* requires further research and specialized tuning to be effective for production use in legal text classification.

Deroy et al. (Deroy et al., 2023) assess the preparation of pre-trained abstract summarization models and LLMs towards summarizing judicial judgments. They use metrics like *ROUGE*, *METEOR*, and *BLEU* to evaluate summary quality compared to expert-written gold standards, focusing on models such as *Text-Davinci-003* and *Turbo-GPT-3.5 (ChatGPT)*. The study identifies significant problems in the summaries, such as the fusion of incorrect sentences, imaginary content, and unnecessary data. These issues are related to the training data and summarization algorithms behind these models. The paper concludes that these models are not yet ready for automatic legal summarization, suggesting that human intervention is required to develop effective error detection approaches and summary techniques in the legal domain.

In (Hargreaves, 2023), LLMs' roles in assisting law professors and offering legal advice have also been studied. Researchers had *ChatGPT* create answers to twenty-four distinct tests from a common law school in an English-speaking country. It discovered that the system worked best on essay-based tests that required students to explain international legislation or basic legal concepts that were not necessarily particular to any country. It

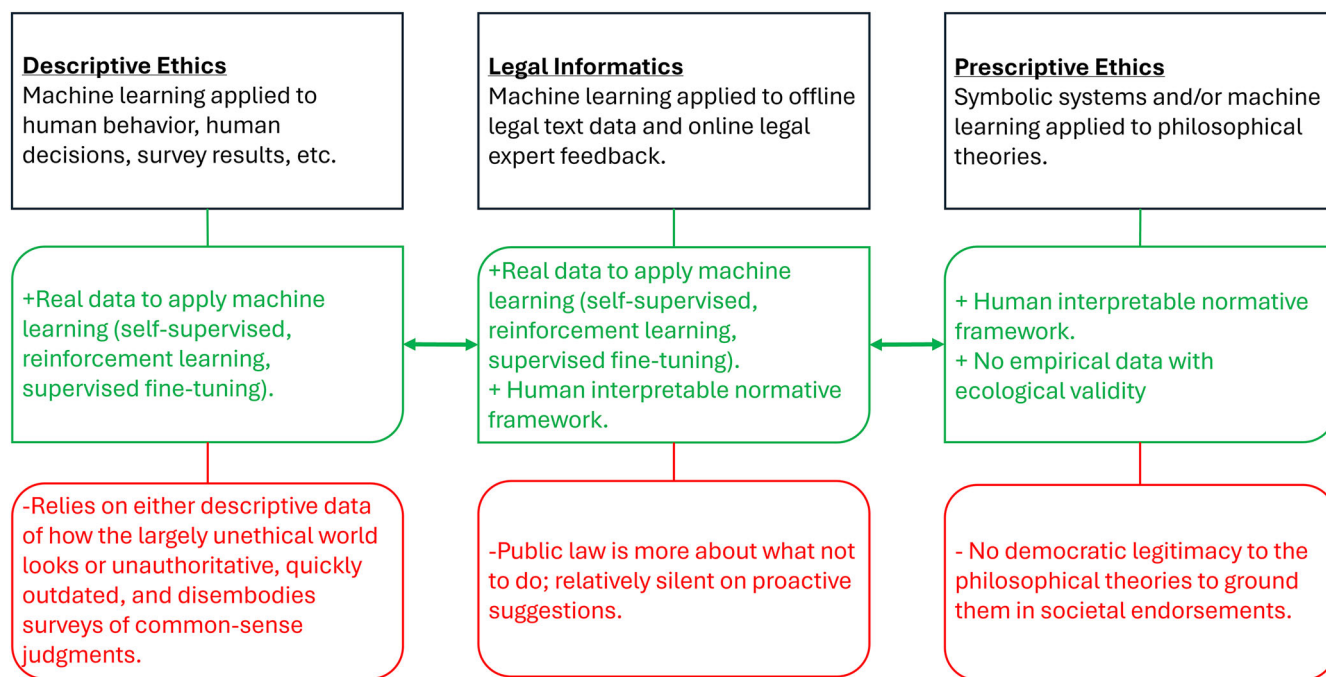


Fig. 7 Comparison of Descriptive Ethics, Legal Informatics, and Prescriptive Ethics in applying machine learning and AI. Descriptive Ethics uses real data but can quickly become outdated and lacks normative guidance. Prescriptive Ethics offers a normative framework but lacks empirical data and democratic legitimacy. Legal Informatics combines real data with a human-interpretable normative framework, making it suitable for aligning AI with societal values (Nay, 2022).

performed the worst on examinations that included “issue spotting” or problem-style questions, where students had to apply local laws or case law to a made-up scenario. They also discussed the moral issues related to using *ChatGPT* and other AI language models for evaluation purposes in law school. Acknowledging the increasing prevalence of AI in educational environments, they put forward perceptive methods to instruct students on the proper and moral application of AI in their coursework.

In the study by (Yu et al., 2022), Chain-of-Thought (CoT) prompts are presented. These prompts help LLMs produce relevant and cohesive phrases with a logical structure, simulating the analytical process of a lawyer. They investigated the effectiveness of these strategies in legal reasoning tasks and used the COLIEE entailment problem, which is modeled after the Japanese Bar exam, to assess zero-shot/few-shot and fine-tune strategies. Their results demonstrate that, while CoT prompts and techniques for fine-tuning with explanations offer advances, prompts originating from certain legal reasoning processes, such as IRAC (Issue, Rule, Application, Conclusion), yield the greatest outcomes. Figure 6 shows the whole process of Zero-Shot (ZS) CoT. They extracted the whole GPT-3 thinking process using the “reasoning” prompt, and they extracted the correct answer from the reasoning text using the “answer” prompt.

There has been discussion about how LLMs could affect the legal profession; this topic is covered in (Iu and Wong, 2023). They compared *ChatGPT*’s proficiency with that of a first-year law student to examine its talents in legal duties. It displays competence in the creation of simple legal papers, providing legal counsel, and formulating reasoned arguments. Despite these advantages, *ChatGPT* is constrained by its incapacity to get up-to-date case laws and carry out exhaustive legal research. The study highlights that although *ChatGPT* is a useful tool for legal work, it cannot completely replace litigation lawyers since human judgment is required for complex legal decisions and current case law interpretation. According to the article, *ChatGPT* should not be considered a stand-alone solution, but rather as an additional instrument in the legal sector.

In (Nay, 2022), the author discusses legal measures towards the alignment of AI with human values. The research tackles the problem of controlling AI activities to fit societal values by advocating for the incorporation of legal procedures into AI systems. The approach is intended to employ clear and flexible legal rules to ensure that AI actions correspond to democratically approved values. Nay points out that AI systems may behave in ways not intended due to misaligned objectives. This can be overcome by embedding legal reasoning into decision-making for AI to help maintain consistency between what is done by human beings and AI systems. It promotes legal informatics as the major way of building AI compatible with human and cultural requirements. Figure 7 shows that Legal informatics does not have the unavoidable drawbacks of both prescriptive and descriptive ethics, but it does have their favorable aspects.

In this research (Savelka et al., 2023), authors look into the issue of the ability of *GPT-4* augmented using legal statutory provision retrieval techniques in explaining legal concepts from statutory provision. They contrast a basic model, where *GPT-4* explains legal terms directly, and an enhanced approach that augments it with case law context. The study observes that even though *GPT-4* generates compelling surface explanations, these are mostly untrue. Augmentation considerably enhances content, diminishes hallucination, boosts factuality, clarity, and relevance. This approach is a useful development in legal education and practice that proves a combination of traditional legal information retrieval with LLMs can be used for a better explanation on legal concepts. Figure 8 shows System Architectures Diagrams. The top section is an actual representation of the baseline applying the LLM. The final part is an explanation of the parts of the augmented information retrieval based architecture.

This research (Simmons and Vasa, 2023) examines the application of LLMs for crime detection using surveillance video descriptions. It demonstrates that when given high-quality manually written descriptions of movies, LLMs can detect and

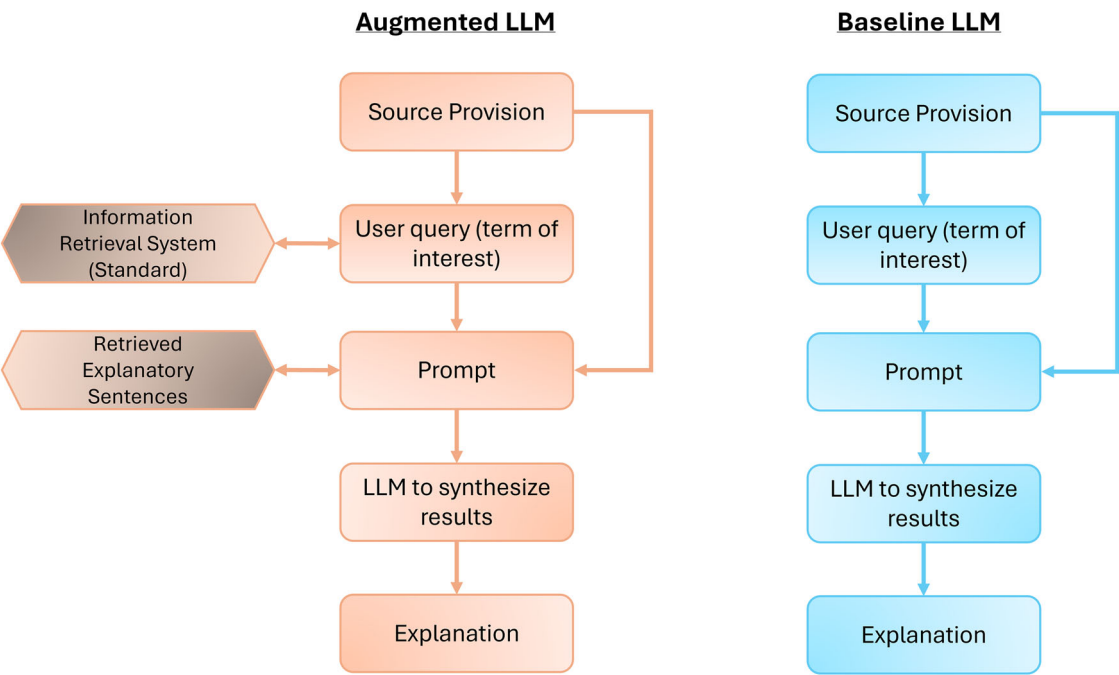


Fig. 8 System Architectures Diagrams comparing the Base-line LLM and the Augmented LLM. The top section represents the baseline LLM process, while the bottom section shows the augmented system incorporating an information retrieval system to provide enhanced explanatory content (Savelka et al., 2023).

Prompt Design

Background Definition:
I will give you the annotation scheme consists of ten data practice categories with its explanation. The annotations are to the website's privacy policy.

Information Instruction:
1. First Party Collection / Use: how and why a service provider collects user information.
9. International & Specific Audiences: practices that pertain only to a specific group of users(e.g., Children, Europeans, or California residents).
10. Other: additional sub-labels for introductory or general text, contact information, and practices not covered by the other categories.

Task Description:
For the text of privacy policy I show you below; please choose the most matching one from these ten categories and tell me. If you don't think there is any matching item, please output 'Other.'

Fig. 9 Description of the prompt layout used in PolicyGPT for analyzing privacy policies. The prompt design includes a background definition, information instruction detailing ten data practice categories, and a task description guiding the selection of the most relevant category for given privacy policy text (Tang et al., 2023).

categorize crimes well utilizing zero-shot reasoning. It does, however, identify fundamental flaws in automatic video-to-text translation approaches, such as a lack of detail, a preference to hallucinate, and an inability to preserve object identification across time. The report emphasizes the importance of domain-specific adjustments in LLMs as well as more research to solve these limits to effective automatic crime detection.

The article (Dal Pont et al., 2023) provides the preliminary findings of the *PRODIGIT* project, a large-scale Italian initiative aiming at assisting tax judges and lawyers using digital technology, with an emphasis on AI. The major goal of the project is to generate judicial decision summaries and extract related data, such as identifying legal concerns, decision-making criteria, and keyword specifications. To achieve these objectives, numerous tools and methodologies for extracting and abstract summarization were deployed and assessed, with a focus on employing LLMs, particularly *GPT-4*. The results were determined to be satisfactory by experienced tax judges and lawyers, leading to the construction of a prototype use that will be publicly available.

According to (Tang et al., 2023), the researchers investigate how LLMs such as *ChatGPT* and *GPT-4* may be used to analyze complicated and verbose privacy regulations. *PolicyGPT*, its creation, intends to make these policies more accessible and comprehensible for users, addressing the widespread issue of people consenting to these policies without fully grasping their contents, which may lead to privacy and legal difficulties. The study illustrates the usefulness of *PolicyGPT* by using it on two significant datasets: *OPP-115*, which contains privacy policies from 115 websites classified into 10 types, and *PPGDPR*, which contains policies from 304 mobile applications annotated with an emphasis on The General Data Protection Regulation (GDPR) user rights. The solid performance of *PolicyGPT*, particularly under zero-shot learning settings, underlines its potential as a means for both professionals and laypeople to better comprehend privacy regulations. This study is noteworthy in the field of legal document analysis because it demonstrates the rising influence of AI and LLMs in simplifying and explaining difficult legal documents for the general public. Figure 9 shows the description of their prompt layout.

In (Zhang et al., 2023b), the authors describe a unique technique for adjusting LLMs such as *GPT-4* to specific domains such as Chinese law, where such models frequently create hallucinatory material due to a lack of in-domain training data.

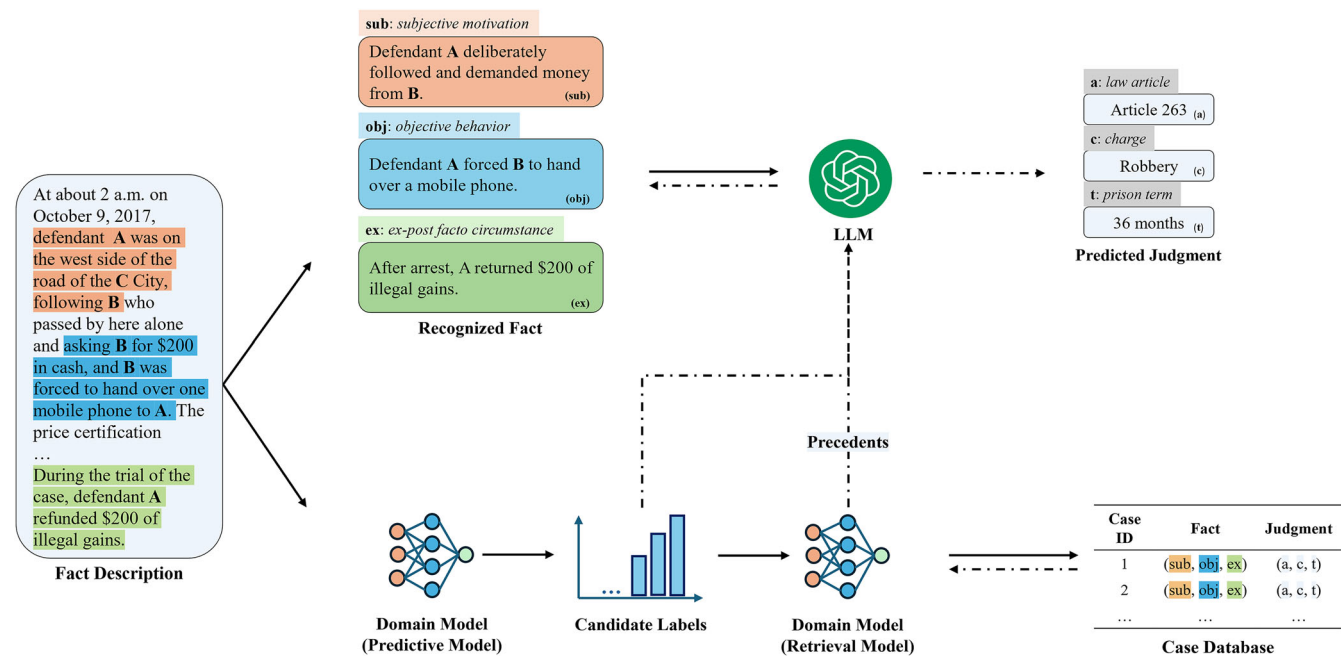


Fig. 10 The complete PLJP framework. The figure illustrates the integration of domain models and LLMs to predict legal judgments. The framework involves extracting case facts (sub: subjective motivation, obj: objective behavior, ex: ex post facto circumstances) and using them to identify relevant precedents and predict judgments (a: law article, c: charge, t: prison term). Solid lines represent the recovery process, while dotted lines indicate the prediction process (Wu et al., 2023b).

The proposed strategy consists of three steps: adapt-retrieve-revise. To begin, a more manageable 7B LLM is fitted to the target domain via continuous learning on in-domain data. This customized LLM provides an initial response to a specific question during task-solving. The draft is then utilized to gather backing proof from an external knowledge source in the domain. Finally, *GPT-4* evaluates and revises the draft response using the recovered evidence to produce the final answer. This methodology intends to capitalize on the smaller model's domain adaption efficiency and *GPT-4*'s capacity to examine evidence, lowering hallucinatory content production. The accuracy of this approach was shown to be 33.3% higher than direct generation by *GPT-4* in a zero-shot condition across four Chinese legal tasks. It also beat two more powerful retrieval-based baselines by 15.4% and 23.9%, respectively.

The integration of LLMs with domain models is proposed in (Wu et al., 2023b) to improve the prediction of legal judgments, which is a key component of legal reasoning. The approach involves selecting potential labels guided by domain models, which help identify relevant legal precedents. These precedents, along with accurately interpreted case facts, are then fed into LLMs to support judge forecasting. In experimental research based on the CAIL2018 dataset, the PLJP framework mimics the judicial procedure of real-life judges. Such an approach represents a significant advancement in legal AI as a method of improving its accuracy and legitimacy of forecasts on court judgment results by combining LLM's broad language understanding capabilities with a highly specialized model. Figure 10 depicts the complete PLJP framework, where the terms sub, obj, and ex relate to subjective motivation, objective behavior, and ex post facto situation, respectively. The solid lines indicate the previous recovery process, whereas the dotted lines reflect the prediction process.

The study (Janatian et al., 2023) analyzes the utilization of LLMs such as *GPT-4* for structuring legislative documents. This method is critical for developing legal expert systems that assist ordinary people in understanding legal norms. The authors assessed the usefulness of LLMs in automation of this procedure

and compared LLM-generated paths to those constructed manually. They discovered that 60% of LLM paths were similar to or better than manual pathways, implying that LLMs can greatly speed up the creation of legal professional systems. This study emphasizes the possible uses of LLMs in the area of legal technology, notably in simplifying legislative text encoding and improving legal decision assistance systems.

According to (Lee, 2023), the researcher introduces *LexGPT* models, which are built upon *GPT-J* models and pre-trained with the Pile of Law dataset. The aim is to create foundational models for future legal applications, emphasizing a "No Code" approach for legal professionals. This approach allows legal experts to use and customize these models without needing extensive technical knowledge. The paper also explores the potential of *LexGPT* as a classifier, acknowledging its current performance limitations and the need for future research to enhance effectiveness without modifying the underlying model or code. *LexGPT* models, designed to facilitate natural language processing in the legal field, will be made publicly available, highlighting their potential use in legal technology.

The authors of this study (Qasem et al., 2023) investigate the installation of a *ChatGPT*-based chatbot for legal help. The chatbot, which was created to assist Palestinian cooperatives, was tested using 50 legal inquiries and demonstrated an accuracy of 82% and an F1-score of 79%. The writers praised the chatbot's ability to accurately answer and cite laws, but they also discovered difficulties with larger legal documents and regulations. This study emphasizes the chatbot's legal help potential, pointing to future enhancements for more complete legal support systems.

The article (Cui et al., 2023) explains the creation of Chat-Law, a specialized legal LLM concentrating on the Chinese legal setting. It highlights the relevance of domain-specific data in decreasing illusions in LLMs and offers an assessment system based on ELO model ranking. This technique, inspired by chess rating systems, objectively examines and compares model performance. The ChatLaw approach uses legal domain

information to enhance legal decision-making and counseling and incorporates vector knowledge databases to boost performance.

The capabilities of LLMs in legal systems heavily depend on the availability and quality of training data. To support these advanced applications, several domain-specific datasets have been developed that capture the complexity and nuance of legal language. The following section reviews key law data resources that are instrumental in training, fine-tuning, and evaluating legal-oriented LLMs.

Law Data Resources for LLMs

LLMs have gained prominence in the legal field due to their potential in tasks such as case retrieval, legal judgment prediction, and understanding legal positions. To effectively adapt and fine-tune these models to the distinctive language and specialized knowledge of law, specific legal datasets are essential. This section explores several pivotal legal datasets, detailing their applications in LLM training and the challenges they present.

- a. *CAIL2018 Dataset*: The CAIL2018 dataset, introduced in (Xiao et al., 2018), comprises over 2.6 million Chinese criminal case documents. It is one of the largest publicly available legal judgment prediction datasets and includes rich annotations on relevant laws, charges, and sentence terms. This enables LLMs to be trained for multi-label classification tasks and legal reasoning generation. However, its focus on the Chinese legal system presents challenges for generalization to other jurisdictions, and the class imbalance in certain charges may cause performance degradation without additional sampling or weighting techniques.
- b. *CaseHOLD Dataset*: The CaseHOLD dataset, presented in (Zheng et al., 2021), includes approximately 53,000 multiple-choice questions derived from United States court rulings. These questions test a model's ability to select the correct legal holding from distractor sentences. Two domain-specific models, *BERT-Law* and *BERT-CaseLaw*, were pretrained using this dataset to improve understanding of legal argumentation and domain-specific vocabulary. Challenges include ambiguity in legal phrasing and the fine semantic distinctions that models must capture to choose the correct holding.
- c. *LeCaRD Dataset*: The LeCaRD dataset (Ma et al., 2021) focuses on legal case retrieval in the Chinese legal system. It contains 43,000 candidate cases and 107 manually selected query cases, each annotated for relevance. The dataset is especially valuable for training retrieval-augmented LLMs and evaluating semantic matching in legal information retrieval. A challenge lies in modeling the unique structure and citation logic of Chinese judicial texts, which differ significantly from Western legal systems.
- d. *MultiLegalPile Dataset*: MultiLegalPile (Niklaus et al., 2023) is a multilingual and multi-jurisdictional dataset consisting of 689GB of legal text across 24 languages and 17 jurisdictions. It aggregates various corpora such as EUR-Lex, Legal multilingual Colossal Clean Crawled Corpus (Legal mC4), and other national legislative texts. This dataset is used for multilingual pretraining and cross-jurisdictional legal text modeling. However, its diverse source licensing and variable data quality present legal and technical challenges, and balancing across under-resourced jurisdictions remains difficult.
- e. *LeXFiles Dataset*: The LeXFiles dataset, introduced by (Chalkidis et al., 2023), is a comprehensive corpus of over 622,000 legal documents from diverse legal systems

including the European Union, United Kingdom, United States, Canada, and India. It includes legislation, case law, and contracts, enabling training of general-purpose legal LLMs across jurisdictions. With over 1.2 billion tokens, LeXFiles supports both classification and generation tasks. One key challenge is ensuring that models handle jurisdiction-specific terminology and contextual differences in law interpretation.

- f. *Cambridge Law Corpus (CLC)*: The Cambridge Law Corpus (CLC) (Östling et al., 2023) consists of over 250,000 court decisions from 53 courts in the United Kingdom, covering a timeline from 1573 to the present. The dataset includes per-case metadata and expert annotations of over 600 legal case studies. It is valuable for historical legal analysis and training LLMs like *GPT-3*, *GPT-4*, or *RoBERTa* in understanding legal evolution. The primary challenge with CLC is managing diachronic language variations, legal shifts over time, and long-document summarization tasks.
- g. *Legal Extractive Summarization Dataset*: This dataset (Bauer et al., 2023) includes 430,000 United States court decisions from 1755 to 2016, annotated with extractive summaries that reflect the most legally relevant portions of the cases. It is used to train summarization models that replicate how lawyers distill lengthy cases into actionable insights. The average summary length is 6 sentences with a compression ratio of 15.8 percent. One challenge lies in aligning the model's focus with human annotator priorities, especially in longer or structurally diverse cases.
- h. *European Court of Human Rights (ECtHR) Dataset*: This dataset (Chalkidis et al., 2019) consists of thousands of cases labeled with one or more articles of the European Convention on Human Rights allegedly violated. It is widely used for tasks such as legal judgment prediction and multi-label classification. This dataset presents opportunities for studying international legal reasoning, but also faces challenges due to the complexity of human rights law and variation in how articles are interpreted across different member states.
- i. *United States Supreme Court (SCOTUS) Dataset*: This dataset (Spaeth et al., 2020) collects decades of Supreme Court decisions, annotated with voting breakdowns, issue areas, and metadata. It is extensively used in the study of judicial behavior, constitutional law modeling, and long-term trend analysis. While valuable for studying high-level legal reasoning, this dataset presents challenges such as limited annual case volume, changes in court composition, and the evolution of legal doctrine over time.
- j. *UNFAIR Terms of Service (UNFAIR-ToS) Dataset*: This dataset (Lippi et al., 2019) comprises Terms of Service agreements from 50 online platforms, annotated for potentially unfair clauses based on European Union consumer protection law. It is highly relevant for training language models to automatically identify manipulative, non-transparent, or legally questionable contractual language. A key challenge in utilizing this dataset is the nuanced nature of legal unfairness, which often depends on specific context, subtle wording, and differing interpretations of consumer rights across jurisdictions.

Table 1 provides a categorized summary of key legal datasets used to train and evaluate LLMs, organized by jurisdictional coverage and task type.

The Implications of LLMs

The increasing use of LLMs in various domains, including legal applications, brings about significant implications that indicate

Table 1 Summary of Law Data Resources for LLMs.			
Dataset Name	Description	Applications	Size
CAIL2018 (Xiao et al., 2018)	Over 2.6 million criminal cases from China, annotated with laws, charges, and sentences.	Legal judgment prediction (LJP), multi-task learning, classification	2.6 million cases
CaseHOLD (Zheng et al., 2021)	53,000 multiple-choice questions based on U.S. court rulings.	Legalspecializationmodels, holding prediction	53,000 questions
LeCaRD (Ma et al., 2021)	Legal case retrieval dataset with 43,000 candidates and 107 queries (China).	Semantic legal retrieval, similarity learning	43,000 candidate cases, 107 query cases
MULTILEGALPILE (Niklaus et al., 2023)	689GB multilingual legal dataset across 24 languages and 17 jurisdictions.	Multilingual legal pretraining, fair use research	689GB
LeXFiles (Chalkidis et al., 2023)	Over 1.2B tokens from legal documents in the EU, US, UK, Canada, and India.	Legal LLM pretraining, classification and generation	Millions of documents
Cambridge Law Corpus (CLC) (O’stling et al., 2023)	250,000 + UK court decisions from 1573 to present, with annotations.	Historical legal analysis, LLM fine-tuning	0.8 billion tokens
Legal Extractive Summarization (Bauer et al., 2023)	430,000 US court decisions annotated with extractive summaries.	Legal summarization, document compression	430,000 court decisions
ECtHR (Chalkidis et al., 2019)	Human rights cases labeled by violated articles of the European Convention.	Multi-label classification, LJP, legal reasoning	Thousands of cases
SCOTUS (Spaeth et al., 2014)	Annotated US Supreme Court decisions with issue tags and voting data.	Judicial behavior modeling, constitutional law trends	Decades of rulings
UNFAIR-ToS (Lippi et al., 2019)	Terms of Service agreements annotated for unfair clauses (EU law).	Consumer protection analysis, unfair contract detection	50 ToS agreements

careful consideration. This section delves into several key issues associated with LLMs, supported by relevant research findings.

A. Bias and Fairness. Research by Felkner et al. examines biases in LLMs, particularly against the LGBTQ+ community, using models like *BERT*. The study introduces an innovative technique for mitigating these biases by training models with data specific to the LGBTQ+ population. They propose a new benchmark dataset, *WinoQueer*, highlighting that biases in off-the-shelf models can be substantially reduced through fine-tuning. This research underscores the importance of incorporating diverse perspectives in AI training to enhance fairness and eliminate biases (Felkner et al., 2022).

Similarly, Abid et al. investigate anti-Muslim prejudice in the *GPT-3* language model. Their findings reveal that *GPT-3* often associates the concept of being Muslim with violence, leading to biased outputs. They explore various methods, such as prompt engineering and analogical reasoning, to identify and mitigate this bias. This study emphasizes the challenges in addressing deep-seated biases in language models and the necessity for continuous monitoring and adjustment (Abid et al., 2021).

In legal contexts, such biases can manifest in various ways. For instance, biased LLMs might overestimate the likelihood of recidivism for specific groups, impacting risk assessments and sentencing recommendations. They might also omit favorable precedents or overemphasize unfavorable ones during legal research, leading to unfair judicial outcomes (David C Donald).

B. Legal and Ethical Considerations. Cheong evaluates whether existing US laws adequately address the threats posed by generative AI to human values. The study points out significant gaps and ambiguities in current legal frameworks, particularly concerning principles of confidentiality, neutrality, inclusion, fairness, and mental health. It argues that even fundamental and civil rights may fall short of protecting against AI-driven

discriminatory outcomes, advocating for proactive regulations that anticipate new risks and provide clear guidelines for industry players (Cheong et al., 2023).

C. Misinformation and Hallucinations. Li et al. critically examine the problems of false information and hallucinations produced by LLMs such as ChatGPT. The study draws attention to the risks associated with these algorithms creating false information and repeating biased or inaccurate data—a phenomena that has been compared to “stochastic parrots” The paper (Li, 2023) criticizes the shortcomings of the European AI regulatory framework in tackling these issues and advocates for stricter regulations and increased user education, particularly in delicate fields like law and healthcare.

Real-world failures have further illustrated these limitations. In one case, lawyers submitted a legal brief containing fictitious case citations generated by ChatGPT, which led to court sanctions. Such hallucinations raise serious concerns about the reliability of LLMs for legal drafting and research (Raymond, 2023). Similarly, in a separate case, an immigration lawyer filed a motion that included six non-existent court decisions produced by an AI tool, resulting in public reprimand and a requirement to complete ethics training (Raymond, 2023).

Impact on the Judicial System

Grossman explores the influence of generative AI technologies on the judicial system. The study identifies several challenges, including:

- **Veracity and Authenticity of AI-Generated Evidence:** One of the primary concerns is the reliability and trustworthiness of evidence generated by AI systems. There are questions about whether AI-generated content can be manipulated or falsified, and how to ensure its integrity. This challenge is crucial because the judicial system relies heavily on the authenticity of evidence presented in court.

- **Intellectual Property Concerns:** Generative AI raises significant intellectual property issues, particularly around authorship and copyright. Determining the ownership of content created by AI can be complex, especially when multiple parties are involved in the development and training of the AI. This complexity can lead to disputes over the rights to use and distribute AI-generated works.
- **Legal Implications in Litigation:** The use of AI in legal decision-making processes brings up various legal implications. There is a concern about the extent to which AI can or should influence judicial decisions. Questions arise about accountability and liability when AI systems are used to aid in or make legal determinations. Additionally, there is the potential for biases in AI algorithms, which could impact the fairness of outcomes in litigation.
- **Need for Adaptive Legal Frameworks:** Current legal frameworks may not be sufficient to address the rapidly evolving capabilities of AI technologies. As AI becomes more integrated into the judicial system, there is a pressing need for laws and regulations to adapt accordingly. This adaptation includes establishing clear guidelines on the use of AI in legal contexts, setting standards for the development and deployment of AI systems, and ensuring that these technologies are used ethically and responsibly.

The research suggests that current legal frameworks may be inadequate to address these new challenges, stressing the necessity for legal systems to evolve alongside AI advancements (Grossman et al., 2023).

Limitations

The use of *ChatGPT* and other LLMs in legal writing and analysis presents several significant limitations. According to Choi et al., one of the primary challenges is that *ChatGPT* is designed as a general-purpose language model and lacks specific training tailored to legal analysis or the construction of test solutions. Its performance in law school examinations has been inconsistent, highlighting its inability to replace human expertise in legal analysis and writing consistently. Furthermore, the integration of AI models like *ChatGPT* into law school examinations raises ethical and academic integrity concerns that must be addressed by legal educators (Choi et al., 2021).

Armstrong discusses additional limitations and obstacles in using *ChatGPT* within the legal area. One critical issue is the potential for inaccuracies and incorrect responses, which raises ethical questions regarding its use in practice. This can lead to violations of client confidentiality and other model rules of professional conduct. Armstrong also notes that *ChatGPT*'s responses may not always be relevant to specific legal contexts and may lack the depth required for complex legal analysis. Moreover, the dynamic nature of law means that the information provided by *ChatGPT* may be outdated, emphasizing the necessity of consulting skilled lawyers and the latest case law (Armstrong, 2023).

Regalia underscores the need for legal practitioners to ensure that AI-generated information complies with professional conduct guidelines, intellectual property laws, and confidentiality requirements. The use of Generative AI (GAI) tools can also introduce ethical dilemmas, such as potential biases or inaccuracies in legal writing. Given that GAI systems may be trained on extensive legal datasets, they might not always reflect the most current legal standards or decisions. Legal professionals must exercise due diligence in verifying the accuracy and relevance of AI-generated content (Regalia, 2023).

Gandhi et al. raise concerns about the transparency and accountability of outputs generated by LLMs. They highlight that such models have not passed qualification exams to practice law,

which further questions their reliability in providing legal advice. The study emphasizes the importance of continuous human oversight to ensure that AI-generated legal information is accurate and ethically sound (Gandhi and Talwar, 2023).

In summary, while *ChatGPT* and similar LLMs offer promising tools for legal writing and analysis, their use is fraught with limitations and challenges. Legal professionals and educators must remain vigilant in addressing these issues to ensure the ethical and effective integration of AI in the legal field.

While this survey aims to provide a comprehensive overview of LLMs and their applications within legal systems, several limitations should be noted.

First, given the rapid pace of development in natural language processing and legal technology, new LLM architectures, datasets, and applications may have emerged after the completion of this study. As such, the coverage of recent advancements may not be exhaustive.

Second, this survey primarily focuses on English-language literature and legal systems. Consequently, developments in non-English jurisdictions or multilingual LLM applications may not be fully captured.

Third, although we examined a wide range of publications, there may be an inherent bias toward highly visible or well-documented models and applications. Some proprietary or unpublished approaches may have been excluded due to limited publicly available information.

Finally, this study does not include experimental evaluation or benchmarking of the reviewed models. Our conclusions are based on existing literature and reported findings, which may carry limitations inherent to those original studies, including potential bias, incomplete evaluation metrics, or lack of longterm deployment results.

Future Directions

The future of LLMs in legal applications holds significant potential but requires addressing several challenges and exploring promising research directions. As LLMs continue to evolve, several areas need further investigation and development:

1. **Bias Mitigation and Fairness:** Future research should investigate adversarial debiasing techniques, fairness-aware training algorithms, and the use of counterfactual data augmentation to address social and legal bias in AI outputs (Binns, 2018; Weidinger et al., 2022). In legal contexts, this may involve training models to avoid overrepresentation of unfavorable case precedents for marginalized communities.
2. **Explainability and Transparency:** Legal AI systems require high levels of interpretability to build trust and ensure accountability in high-stakes decision-making. Two widely used model-agnostic techniques are SHAP (SHapley Additive exPlanations) and LIME (Local Inter-pretable Model-agnostic Explanations). SHAP leverages concepts from cooperative game theory to compute the contribution of each input feature to the model's output by assigning Shapley values, which offer a globally consistent and locally accurate explanation (Lundberg and Lee, 2017). In contrast, LIME works by perturbing input data and fitting a simple, interpretable surrogate model (like a linear model) to approximate the local behavior of the black-box model near a specific prediction (Ribeiro et al., 2016). These techniques are increasingly being adapted to legal contexts. For example, SHAP can help users understand which statutes or facts most influenced a legal judgment prediction, while LIME can explain why a case was retrieved in response to a legal query. Additionally, emerging techniques such as counterfactual explanations

—where models generate “what-if” scenarios to illustrate decision boundaries—and attention visualization in transformer-based models (e.g., BERT or GPT variants) are being explored for their potential to provide insight into legal reasoning processes (Wachter et al., 2017; Vig, 2019). These methods collectively aid in validating model decisions and increasing transparency in AI-generated legal outcomes (Medvedeva et al., 2020; Mittelstadt et al., 2016).

3. **Prompt Engineering in Legal Contexts:** Unlike general domains, legal prompt engineering must deal with formal language, case citations, and jurisdictional specificity. Techniques such as Legal Prompt Engineering (LPE) and Chain-of-Thought (CoT) prompts structured using legal reasoning templates (e.g., IRAC) are showing promise. Comparative studies on their performance across jurisdictions are needed.
4. **Domain-Specific Training:** Fine-tuning on legal-specific corpora (e.g., statutes, case law, international regulations) are essential. Collaboration between AI developers and legal scholars is vital to create balanced and representative datasets (Surden, 2018).
5. **Ethical and Legal Frameworks:** There is an urgent need for robust governance mechanisms. Policymakers and researchers must craft AI regulations to address issues of accountability, transparency, authorship, and user rights in legal decision-making tools (Castillo-González, 2023; Gandhi and Talwar, 2023).
6. **Human-AI Collaboration:** Future tools should enhance productivity by acting as co-pilots rather than replacements. Interfaces must be developed to allow legal professionals to review, refine, or reject suggestions from LLMs in real time, ensuring human accountability (Choi et al., 2021).
7. **Real-World Validation:** Few legal LLMs have been evaluated in actual law firm or court settings. Future research should include field studies to assess accuracy, usability, and failure modes under live legal conditions (Agarwal et al., 2018).
8. **Multilingual and Cross-Jurisdictional Models:** Legal NLP should extend beyond English. Tools like *LaMDA* and *BLOOM* show promise in supporting multilingual applications, but more effort is needed in aligning legal terminologies across languages.
9. **Continuous Learning and Adaptation:** Legal standards evolve, so LLMs must be continuously updated. Retrieval-Augmented Generation (RAG) pipelines and continual learning can help models stay current with case law and regulations (Surden, 2021; Weidinger et al., 2021).
10. **Future Research Gaps:** The literature remains sparse in several high-impact areas:
 - Real-time evaluation of LLMs in live legal workflows (e.g., court submissions, client consultations).
 - Development and benchmarking of multilingual LLMs for underrepresented legal systems.
 - Long-term frameworks for ethical auditing and bias monitoring in deployed legal AI systems.
 - Evaluating hallucination mitigation strategies, including rule-grounded generation and fact verification pipelines.

Conclusion

In conclusion, this paper provided a comprehensive survey of the applications, innovations, implications, and limitations of LLMs in the legal domain. We explored how LLMs are reshaping tasks such as legal drafting, research, reasoning, education, and client interaction.

Through a detailed review of recent studies, we identified emerging models and prompt engineering techniques tailored to legal contexts, highlighting their capabilities and shortcomings.

Despite these advancements, LLMs still face notable challenges. Ethical concerns such as systemic bias, hallucination, and misinformation remain critical. Real-world incidents—like the use of fabricated case citations—demonstrate the dangers of unpervised reliance on these tools. Additionally, many models lack jurisdiction-specific knowledge, multilingual support, or clear explanations behind outputs, raising questions about fairness, transparency, and legal compliance.

Data availability

This article is a survey study based on previously published literature. No new data were generated or analyzed; hence, data sharing is not applicable.

Code availability

Not applicable.

Received: 3 January 2025; Accepted: 3 September 2025;

Published online: 24 December 2025

References

- Abid A, Farooqi M & Zou J (2021) Persistent anti-muslim bias in large language models. In Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society, pages 298–306
- AIMultiple Research Team. Generative ai legal use cases & examples in 2025. (2025) <https://research.aimultiple.com/generative-ai-legal/>
- Armstrong AB (2023) Who's afraid of chatgpt? an examination of chatgpt's implications for legal writing. An Examination of ChatGPT's Implications for Legal Writing (January 23, 2023)
- Backlinko, B (2024) <https://backlinko.com>. Accessed: 2024-08-01
- Bauer, E, Stambach, D, Gu N & Ash E (2023) Legal extractive summarization of us court opinions. arXiv preprint arXiv:2305.08428
- Binns, R (2018) Fairness in machine learning: Lessons from political philosophy. In Conference on fairness, accountability and transparency, pages 149–159. PMLR
- Blair-Stanek, A, Holzenberger, N & Van Durme, B (2023) Can gpt-3 perform statutory reasoning? arXiv preprint arXiv:2302.06100 <https://arxiv.org/abs/2302.06100>
- Bommarito II, M & Katz, DM (2022) Gpt takes the bar exam. arXiv preprint arXiv:2212.14402
- Brown T, Mann B, Ryder N, Subbiah M, Kaplan JD, Dhariwal P, Neelakantan A, Shyam P, Sastry G, Askell A et al. (2020) Language models are few-shot learners. Adv. neural Inf. Process. Syst. 33:1877–1901
- Castillo González, W (2023) The importance of human supervision in the use of chatgpt as a support tool in scientific writing. Metaverse Basic and Applied Research, (2):5
- Chalkidis, I (2023) Chatgpt may pass the bar exam soon, but has a long way to go for the lexglue benchmark. arXiv preprint arXiv:2304.12202
- Chalkidis, I, Androutsopoulos, I & Aletras, N (2019) Neural legal judgment prediction in english. arXiv preprint arXiv:1906.02059
- Chalkidis, I, Garneau, N, Goanta, C, Katz, D M & Sogaard, A (2023) Lexfiles and legallama: Facilitating english multinational legal language model development. arXiv preprint arXiv:2305.07507
- Chen, H, Zhang, L, Liu, Y, Chen, F & Yu, Y (2022) Knowledge is power: understanding causality makes legal judgment prediction models more generalizable and robust. arXiv preprint arXiv:2211.03046
- Cheong, I, Caliskan, A & Kohno, T (2023) Is the us legal system ready for ai's challenges to human values? arXiv preprint arXiv:2308.15906
- Choi JH, Hickman KE, Monahan AB, Schwarcz D (2021) Chatgpt goes to law school. J. Leg. Educ. 71:387
- Cui J, Li Z, Yan Y, Chen B & Yuan L (2023) Chatlaw: Open-source legal large language model with integrated external knowledge bases. CoRR
- Cui J, Ning M, Li Z, Chen B, Yan Y, Li H, Ling B, Tian Y & Yuan L (2023) Chatlaw: A multi-agent collaborative legal assistant with knowledge graph enhanced mixture-of-experts large language model. arXiv preprint arXiv:2306.16092
- Daniel Mittelstadt B, Allo P, Taddeo M, Wachter S, Floridi L (2016) The ethics of algorithms: Mapping the debate. Big Data Soc. 3(2):2053951716679679
- David C Donald Bias in al large language models: Risks and remedies

- Deroy A, Ghosh, K & Ghosh, S (2023) How ready are pre-trained abstractive models and llms for legal case judgement summarization? arXiv preprint arXiv:2306.01248
- Dropbox. Dropbox announces fiscal 2024 first quarter results, 2024. URL <https://dropbox.gcs-web.com/news-releases/news-release-details/dropbox-announces-fiscal-2024-first-quarter-results>. Accessed: 2024-08-01
- Du, Y, Watkins, O, Wang, Z, Colas, C, Darrell, T, Abbeel, P, Gupta A & Andreas, J. (2023) Guiding pretraining in reinforcement learning with large language models. arXiv preprint arXiv:2302.06692
- Duarte F, Number of chatgpt users (2023) <https://explodingtopics.com/blog/chatgpt-users>, 2023. Accessed: 2023-11-03
- de Faria JR, Xie H & Steffek F (2024) Automatic information extraction from employment tribunal judgements using large language models. arXiv preprint arXiv:2403.12936
- Felkner VK, Herbert Chang H-C, Jang E & May J (2022) Towards winoquer: Developing a benchmark for anti-queer bias in large language models. arXiv preprint arXiv:2206.11484
- Foursquare (2024) Foursquare statistics <https://99firms.com/research/foursquare-statistics/>. Accessed: 2024-08-01
- Gandhi P, Talwar V (2023) Artificial intelligence and chatgpt in the legal context. Indian J. Med. Sci. 75(1):1–2
- Grossman MR, Grimm PW, Brown DG, Xu M (2023) The gptjudge: Justice in a generative ai world. Duke L. Tech. Rev. 23:1
- Hargreaves S (2023) 'words are flowing out like endless rain into a paper cup': Chatgpt & law school assessments. Leg. Educ. Rev. 33:69
- Huang Q, Tao M, Zhang C, An Z, Jiang C, Chen Z, Wu Z & Feng Y (2023) Lawyer llama technical report. arXiv preprint arXiv:2305.15062
- Iu, K Y & Man-Yi Wong, V (2023) Chatgpt by openai: The end of litigation lawyers? Available at SSRN 4339839
- Janatani, S, Westermann, H, Tan, J, Savelka, J & Benyekhle, K (2023) From text to structure: Using large language models to support the development of legal expert systems. In Legal Knowledge and Information Systems, pages 167–176. IOS Press
- Jiang, C & Yang X (2023) Legal syllogism prompting: Teaching large language models for legal judgment prediction. In *Proc of the nineteenth international conference on artificial intelligence and law* 417–421
- Jieh-Sheng Lee (2023) Lexgpt 0.1: pre-trained gpt-j models with pile of law. arXiv preprint arXiv:2306.05431
- Kickstarter. About kickstarter, 2024. <https://www.kickstarter.com/about>. Accessed: 2024-08-01
- Lee K, Firat O, Agarwal A, Fannjiang C & Sussillo D (2018) Hallucinations in neural machine translation
- Li, Z (2023) The dark side of chatgpt: Legal and ethical challenges from stochastic parrots and hallucination. arXiv preprint arXiv:2304.14347
- Lippi M, Palka P, Contissa G, Lagioia F, Micklitz H-W, Sartor G, Torroni P (2019) Claudette: an automated detector of potentially unfair clauses in online terms of service. Artif. Intell. Law 27(2):117–139
- Lundberg, SM & Lee S-I (2017) A unified approach to interpreting model predictions. Advances in neural information processing systems, 30
- Ma, Y, Shao, Y, Wu, Y, Liu, Y, Zhang, R, Zhang, M & Ma S (2021) Lecard: a legal case retrieval dataset for chinese law system. In *Proc of the 44th international ACM SIGIR conference on research and development in information retrieval* 2342–2348
- Macey-Dare R (2023) How chatgpt and generative ai systems will revolutionize legal services and the legal profession. Available at SSRN 4366749
- Magesh V, Surani F, Dahl M, Suzgun M, Manning CD & Ho DE (2024) Ai on trial: Legal models hallucinate in 1 out of 6 (or more) benchmarking queries
- Martin Katz D, James Bommarito M, Gao S, Arredondo P (2024) Gpt-4 passes the bar exam. Philos. Trans. R. Soc. A 382(2270):20230254
- Medvedeva M, Vols M, Wieling M (2020) Using machine learning to predict decisions of the european court of human rights. Artif. Intell. Law 28(2):237–266
- Nay JJ (2022) Law informs code: A legal informatics approach to aligning artificial intelligence with humans. Nw. J. Tech. Intell. Prop. 20:309
- Nay J J (2023) Large language models as fiduciaries: a case study toward robustly communicating with artificial intelligence through legal standards. arXiv preprint arXiv:2301.10095
- Nexis, L (2023) generative ai the legal profession. <https://cert-www.lexisnexis.com/pdf/lexisnexis-legal-ai-sample-report.pdf>, 2023. Accessed: 2023-03-15
- Niklaus J, Chalkidis, I & Stürmer, M (2021) Swiss-judgment-prediction: A multilingual legal judgment prediction benchmark. arXiv preprint arXiv:2110.00806
- Niklaus J, Matoshi V, Stürmer M, Chalkidis I & Ho, D E (2023) Multilegalpile: A 689gb multilingual legal corpus. arXiv preprint arXiv:2306.02069
- O' stling, A, Sargeant H, Xie H, Bull L, Terenin A, Jonsson L, Magnusson M & Steffek F (2024) The cambridge law corpus: a corpus for legal ai research. Adv Neural Inf Process Syst, 36
- Oltz TP (2023) Chatgpt, professor of law. U. Ill. JL Tech. & Pol'y, page 207
- Parizi AH, Liu Y, Nokku P, Gholamian S, Emerson D (2023) A comparative study of prompting strategies for legal text classification. Proc. Nat. Leg. Lang. Process. Workshop 2023:258–265. pages
- Perlman A (2023) The implications of chatgpt for legal services and society. Mich. Tech. L. Rev. 30:1
- Pont T D, Galli F, Loreggia A, Pisano G, Rovatti R & Sartor, G (2023) Legal summarisation through llms: The prodigat project. arXiv preprint arXiv:2308.04416
- Qasem R, Tantour B & Maree M (2023) Towards the exploitation of llm-based chatbot for providing legal support to palestinian cooperatives. arXiv preprint arXiv:2306.05827
- Qin W & Sun Z. (2024) Exploring the nexus of large language models and legal systems: A short survey. arXiv preprint arXiv:2404.00990
- Raymond N (2023) New york lawyers sanctioned for using fake chatgpt cases in legal brief. <https://www.reuters.com/legal/new-york-lawyers-sanctioned-using-fake-chatgpt-cases-legal-brief-2023-06-22/>, Accessed: 2025-04-12
- Regalia, J. Chatgpt and legal writing: The perfect union? SSRN Electronic Journal, 2023. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4371460
- Ribeiro MT, Singh S, Guestrin C (2016) Why should i trust you? Explaining the predictions of any classifier. In *Proc of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*. 1135–1144
- Savelka J, Ashley K D, Gray M A, Westermann H & Xu H (2023) Explaining legal concepts with augmented large language models (gpt-4). arXiv preprint arXiv:2306.09525
- SEO.ai. How many users does chatgpt have?, 2024. URL <https://seo.ai/blog/how-many-users-does-chatgpt-have>. Accessed: 2024-08-01
- Shao, Z, Yu, Z, Wang, M & Yu, J (2023) Prompting large language models with answer heuristics for knowledgebased visual question answering. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 14974–14983
- Shui, R, Cao, Y, Wang, X & Chua, T-S (2023) A comprehensive evaluation of large language models on legal judgment prediction. pages 7337–7348, <https://doi.org/10.48550/arXiv.2310.11761>
- Simmons A & Vasa R (2023) Garbage in, garbage out: Zero-shot detection of crime using large language models. arXiv preprint arXiv:2307.06844
- Spaeth H, Epstein L, Ruger T, Whittington K, Segal J & Martin, AD (2014) Supreme court database code book. URL: <http://scdb.wustl.edu>, 663
- Statista. Quarterly number of netflix streaming subscribers worldwide, 2024. URL <https://www.statista.com/statistics/250934/quarterly-number-of-netflix-streaming-subscribers/>. Accessed: 2024-08-01
- Strickson, B & De La Iglesia, B (2020) Legal judgement prediction for uk courts. In Proceedings of the 3rd International Conference on Information Science and Systems, pages 204–209
- SuperAnnotate. What are large language models (llms): Complete overview 2023, 2023. URL <https://www.superannotate.com/blog/llm-overview>
- Surden H (2018) Artificial intelligence and law: An overview. Ga. St. UL Rev. 35:1305
- Takahashi K, Yamamoto K, Kuchiba A, Koyama T (2022) Confidence interval for micro-averaged f1 and macro-averaged f1 scores. Appl. Intell. 52(5):4961–4972
- Tang C, Liu Z, Ma C, Wu Z, Li Y, Liu W, Zhu D, Li Q, Li X, Liu T et al. (2023) Policygpt: Automated analysis of privacy policies with large language models. arXiv preprint arXiv:2309.10238
- Touvron H, Lavril T, Izacard G, Martinet X, Lachaux M-A, Lacroix T, Rozière B, Goyal N, Hambro E, Azhar F et al. (2023) Llama: Open and efficient foundation language models. arXiv preprint arXiv:2302.13971
- Trautmann, D, Petrova, A & Schilder, F (2022) Legal prompt engineering for multilingual legal judgement prediction. arXiv preprint arXiv:2212.02199
- Tulio Ribeiro, M, Singh, S & Guestrin, C (2016) why should i trust you? explaining the predictions of any classifier. In Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining, pages 1135–1144
- Vig, J & Belinkov, Y Analyzing the structure of attention in a transformer language model. arXiv preprint arXiv:1906.04284, 2019
- Wachter S, Mittelstadt B, Russell C (2017) Counterfactual explanations without opening the black box: Automated decisions and the gdpr. Harv. JL Tech. 31:841
- Warren, Z Generative ai in law firms: For many, such technologies are still a great unknown. https://www.reuters.com/legal/transactional/generative-ai-law-firms-many-such-technologies-are-still-great-unknown-2023-05-23/?utm_source=chatgpt.com, May 2023. Accessed: 2023-05-23
- Weichbroth P (2025) Ai and the law: Evaluating chatgpt's performance in legal classification. arXiv preprint arXiv:2502.12193
- Weidinger L, Uesato J, Rauh M, Griffin C, Huang PS, Mellor J, Glaese A, Cheng M, Balle B, Kasirzadeh A, et al. (2022) Taxonomy of risks posed by language models. In *Proc of the 2022 ACM conference on fairness, accountability, and transparency* 214–229
- Workshop BS, Scao TL, Fan A, Akiki C, Pavlick E, Iliæ S, Hesslow D, Castagné R, Luccioni AS, Yvon F, et al. (2022) Bloom: A 176b-parameter open-access multilingual language model. arXiv preprint arXiv:2211.05100
- Wu Y, Zhou S, Liu Y, Lu W, Liu X, Zhang Y, Sun C, Wu F & Kuang K (2023b) Precedent-enhanced legal judgment prediction with llm and domain-model collaboration. arXiv preprint arXiv:2310.09241
- Wu, C, Yin, S, Qi, W, Wang, X, Tang, Z & Visual ND (2023a) chatgpt: Talking, drawing and editing with visual foundation models. arXiv preprint arXiv:2303.04671

- X Zang and Others (2024) The implications of chatgpt for legal services and society <https://clp.law.harvard.edu/article/the-implications-of-chatgpt-for-legal-services-and-society>, Accessed: YYYY-MM-DD
- Xiao C, Zhong H, Guo Z, Tu C, Liu Z, Sun M, Feng Y, Han X, Hu Z, Wang H, et al. Cail2018: A large-scale legal dataset for judgment prediction. arXiv preprint arXiv:1807.02478, 2018
- Yu F, Quartey, L & Schilder, F (2022) Legal prompting: Teaching a language model to think like a lawyer. arXiv preprint arXiv:2212.01326
- Yue S, Chen W, Wang S, Li B, Shen C, Liu S, Zhou Y, Xiao Y, Yun S, Huang X, et al. (2023) Disc-lawllm: Fine-tuning large language models for intelligent legal services. arXiv preprint arXiv:2309.11325
- Zhang H, Dou Z, Zhu Y, Wen J-R (2023a) Contrastive learning for legal judgment prediction. ACM Trans. Inf. Syst. 41(4):1–25
- Zhang Y, Wang Y, Cheng F, Kurohashi S, et al. (2023b) Reformulating domain adaptation of large language models as adapt-retrieve-revise: A case study on chinese legal domain. arXiv preprint arXiv:2310.03328
- Zheng L, Guha N, Anderson BR, Henderson P & Ho DE (2021) When does pretraining help? assessing self-supervised learning for law and the casehold dataset of 53,000+ legal holdings. In *Proc of the eighteenth international conference on artificial intelligence and law* 159–168

Acknowledgements

Not applicable.

Author contributions

Author1, Author2, and Author3 contributed equally to the writing of the manuscript. The last author supervised the project and provided overall guidance for improving the manuscript. Author1 served as the corresponding author. Competing interests
The authors declare that they have no conflicts of interest or competing interests.

Ethical Approval

This article does not contain any studies involving human participants or animals performed by any of the authors.

Informed Consent

Not applicable, as this study did not involve human participants.

Additional information

Correspondence and requests for materials should be addressed to Fatemeh Dehghani.

Reprints and permission information is available at <http://www.nature.com/reprints>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.



Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

© The Author(s) 2025